

Multiple Choice (10 marks)

1. Besides temperature, what other physical conditions must be taken into account for the growth of bacteria?
 - (a) Levels of oxygen, acidity or alkalinity of the medium
 - (b) Temperature changes, light exposure
 - (c) Atmospheric pressure, noise levels
 - (d) Nitrogen levels, radiation exposure
 - (e) Soil texture, light wavelength
2. Two individuals, one with type B blood and one with type AB blood, have a child. The probability that the child has type O blood is
 - (a) 50%
 - (b) 100%
 - (c) 0%
 - (d) 10%
 - (e) 75%
3. What functions are served by sound communication in animals?
 - (a) Mate attraction, species recognition, defense, and territorial claim
 - (b) For heat regulation
 - (c) For hunting
 - (d) For navigation
 - (e) For mineral absorption
4. Targeting of a newly synthesized protein is most likely to require two different signal peptides for which of the following destinations?
 - (a) Ribosome
 - (b) Mitochondrion
 - (c) Chloroplast
 - (d) Lysosome
 - (e) Endoplasmic reticulum
5. The amino acid ornithine is regenerated during each turn of the urea cycle in a similar manner to which of the following participants in the citric acid cycle?

- (a) Alpha-ketoglutarate
- (b) Glutamate
- (c) Acetyl CoA
- (d) Oxaloacetate
- (e) Citrate

True / False (10 marks)

Answer True or False

- 6. True or False: Menopausal hormone therapy and irregular endometrial bleeding: a potential role for uterine natural killer cells?
- 7. True or False: Dose end-tidal carbon dioxide measurement correlate with arterial carbon dioxide in extremely low birth weight infants in the first week of life?
- 8. True or False: Do people who were passive smokers during childhood have increased risk of long-term work disability?
- 9. True or False: Profiling quality of care: Is there a role for peer review?
- 10. True or False: Can third trimester ultrasound predict the presentation of the first twin at delivery?

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

- 11. Discuss and analyze the diverse types of evidence from living organisms that support the theory of evolution, emphasizing their significance and interconnections in the evolutionary framework.
- 12. Discuss the mechanisms driving the high rates of leaching and nutrient cycling in tropical rainforest soils, and analyze their ecological implications compared to other biomes.

End of Paper